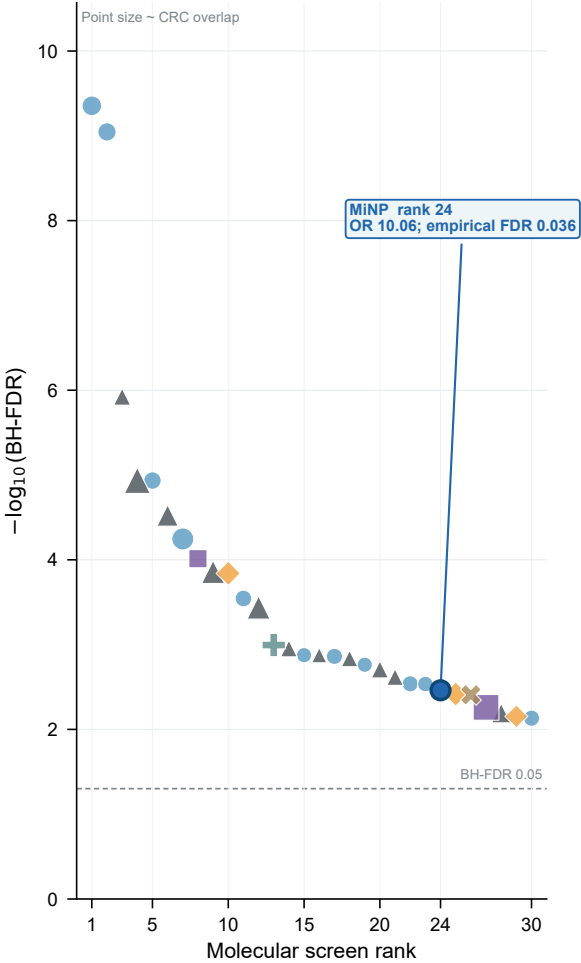
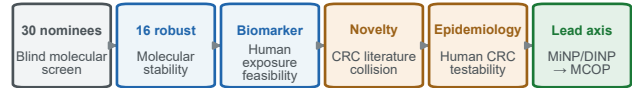


Multistage prioritization advances a rank-24 MiNP signal to a human-testable DINP exposure axis

A Top-30 molecular landscape



B Candidate disposition across four gates



Molecular rank alone did not determine advancement; downstream gates are actionability filters

C Representative candidate evidence matrix

	Molecular signal	Stable	Human biomarker	Multi-cycle	Novelty hits*	Human CRC gate
1	BBzP OR 5.95	Yes	MBzP proxy	10	0	Null†
2	MBzP OR 10.18	Yes	MBzP	10	0	Null
3	Nickel monoxide OR 6.03	Yes	—	—	3	NE
7	MEHP OR 4.34	Yes	MEHP	10	4	Null
8	GenX OR 4.05	Yes	—	—	0	NE
18	Gallium arsenide OR 2.88	Yes	—	—	0	NE
23	TBPH OR 2.83	Yes	—	—	Not audited	NE
24	MiNP/DINP OR 10.06	Yes	MCOP	7	0	Positive

*Exact Title/Abstract query audit (23 Aug 2026); not exhaustive. †BBzP represented by urinary MBzP. NE = not evaluated.

D DINP-axis biomarker translation

